

FITBUDDY

AI FITNESS PLAN GENERATOR

PROJECT DOCUMENTATION

TEAM PROJECT

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1. Abstract

FitBuddy is an AI-based fitness plan generator that helps users create personalized fitness plans. The system collects basic user information such as age, gender, height, weight, fitness goal, and activity level. Using Google Gemini AI, it generates suitable workout and fitness recommendations.

The main purpose of FitBuddy is to provide a simple and personalized fitness planning experience. It can help users understand their fitness goals and follow a structured plan.

2. Introduction

Fitness is an important part of a healthy lifestyle. Many people find it difficult to create a proper workout plan according to their personal needs. FitBuddy is designed to solve this problem using Artificial Intelligence.

The application takes user details as input and uses Gemini AI to generate a personalized fitness plan. The plan can include exercises, workout duration, frequency, and general fitness suggestions.

3. Problem Statement

Many people do not know how to create a suitable fitness plan. Different people have different fitness goals and requirements.

Common problems:

- Lack of personalized workout plans.
- Difficulty in selecting suitable exercises.
- Lack of proper fitness guidance.
- Time required to create a workout schedule.
- Difficulty in maintaining a regular fitness routine.

FitBuddy provides an AI-based solution to these problems.

4. Objectives

- To create personalized fitness plans.
- To use AI for generating workout recommendations.
- To provide simple and easy-to-understand fitness suggestions.
- To save time in creating workout plans.
- To help users follow a structured fitness routine.
- To provide an easy-to-use interface.

5. Existing System

In the existing system, users generally depend on personal trainers, general fitness applications, online workout videos, websites, and fitness articles. These methods may not always provide a plan specifically based on the user's personal information and goals.

6. Proposed System

FitBuddy is an AI-powered fitness plan generator. Users enter their basic details and fitness goals. The system sends the information to Gemini AI and generates a personalized fitness plan.

The proposed system provides:

- Personalized workout suggestions.
- Fitness goal-based recommendations.

- Simple workout schedules.
- AI-generated fitness guidance.
- User-friendly interface.

7. System Requirements

Hardware Requirements:

- Computer/Laptop
- Minimum 4 GB RAM
- Internet connection
- Keyboard and mouse

Software Requirements:

- Python
- FastAPI
- HTML, CSS and JavaScript
- Google Gemini API
- Web browser
- Visual Studio Code

8. System Design

The system consists of three main stages:

User Input → FitBuddy Web Interface → Gemini AI → Generated Fitness Plan

The user enters personal information and fitness goals. The system processes the information and sends a suitable prompt to Gemini AI. Gemini generates the fitness plan, which is then displayed to the user.

9. System Architecture

Architecture Flow:

```
graph TD
    User --> FitBuddy[FitBuddy Web Interface]
    FitBuddy --> Input[User Details & Fitness Goal]
    Input --> Gemini[Gemini AI API]
    Gemini --> Plan[AI Generated Fitness Plan]
    Plan --> Display[Display Plan to User]
```

10. Modules

10.1 User Input Module

This module collects information from the user such as age, gender, height, weight, fitness goal, and activity level.

10.2 AI Fitness Plan Generation Module

This module sends the user's information to Gemini AI. The AI analyzes the input and generates a suitable fitness plan.

10.3 Workout Plan Module

This module displays recommended exercises and workout schedules.

10.4 Feedback Module

Users can submit feedback about their generated plan. The system can use the feedback to update the workout plan.

10.5 Plan Display Module

This module displays the generated plan in a simple and readable format.

11. Technology Used

HTML is used to create the structure of the web application.

CSS is used to design the interface.

JavaScript is used for web interactions.

Python and FastAPI are used for the backend.

Google Gemini AI is used to generate personalized fitness plans.

Swagger UI is used to test and document the API endpoints.

12. Implementation

The implementation starts by creating a web interface for collecting user information. The backend is developed using Python and FastAPI. The user's information is converted into an AI prompt and sent to the Gemini model.

The generated response is processed and displayed as a structured 7-day fitness plan.

The application also supports feedback. Users can request changes to their plan, and the system generates an updated plan based on the feedback.

13. Testing

Testing is performed to verify that the application works correctly.

- Valid user details → Fitness plan generated.
- Different fitness goals → Different plan generated.
- Empty fields → Validation message.
- Valid API request → AI response displayed.
- Feedback submission → Updated plan generated.
- API endpoints → Tested using Swagger UI.

14. Results / Output

The FitBuddy application successfully generates AI-based fitness plans based on user inputs. The generated output contains a 7-day workout schedule, exercise duration, instructions, recovery suggestions, and a nutrition/recovery tip.

The application also allows users to submit feedback and receive an updated workout plan.

15. Advantages

- Easy to use.
- Personalized fitness plans.
- Saves time.
- AI-powered recommendations.
- Simple user interface.

- Supports feedback-based plan updates.
- Easy to modify and improve.

16. Limitations

- Requires an internet connection.
- Requires access to the Gemini API.
- AI-generated suggestions may not be suitable for every individual.
- It is not a replacement for professional medical or fitness advice.

17. Future Enhancement

- User login and profile management.
- Progress tracking.
- Workout reminders.
- Exercise videos.
- BMI calculation.
- Fitness progress charts.
- Voice-based interaction.
- Mobile application support.

18. Conclusion

FitBuddy is an AI-based fitness plan generator designed to make fitness planning easier. It uses user information and Gemini AI to generate personalized workout recommendations.

The project demonstrates how Artificial Intelligence can be used to create useful and interactive applications. With additional features, FitBuddy can be developed into a more complete fitness management application.

19. References

1. Google Gemini AI Documentation
2. FastAPI Documentation
3. HTML Documentation
4. CSS Documentation
5. JavaScript Documentation
6. Google Developers Documentation

20. Database Design / Data Storage Evidence

The supplied project document does not provide a database schema or ER diagram. It does, however, document a “View All Users” screen showing stored workout-plan information. Therefore, this section records the available database evidence without inventing table or field definitions.

Available evidence: The View All Users screen demonstrates that workout-plan information is stored and can be viewed through the application.

Important: The detailed database schema/ER diagram should be added here if the team has an approved database design from the Project Design phase.

21. Installation / Setup Information

The supplied project document identifies the required environment as a computer/laptop with internet access and the following software: Python, FastAPI, HTML, CSS, JavaScript, Google Gemini API, a web browser, and Visual Studio Code.

The source document does not contain exact installation commands or a step-by-step setup procedure. The project's requirements.txt and env.example should be included in the GitHub repository as supporting setup files. A real .env file containing API keys or other secrets should not be uploaded.

22. Screenshots / Output

The following screenshots are retained from the existing FitBuddy project documentation and show the project structure, API endpoints, generated plan, feedback, updated plan, stored information, and nutrition/recovery output.

20. Screenshots / Output

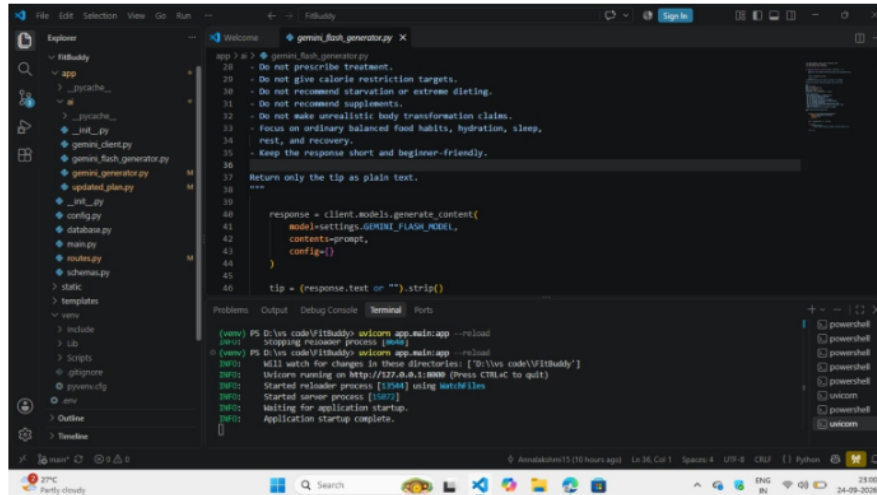


Figure 1: Visual Studio Code project structure and Gemini AI generator code.

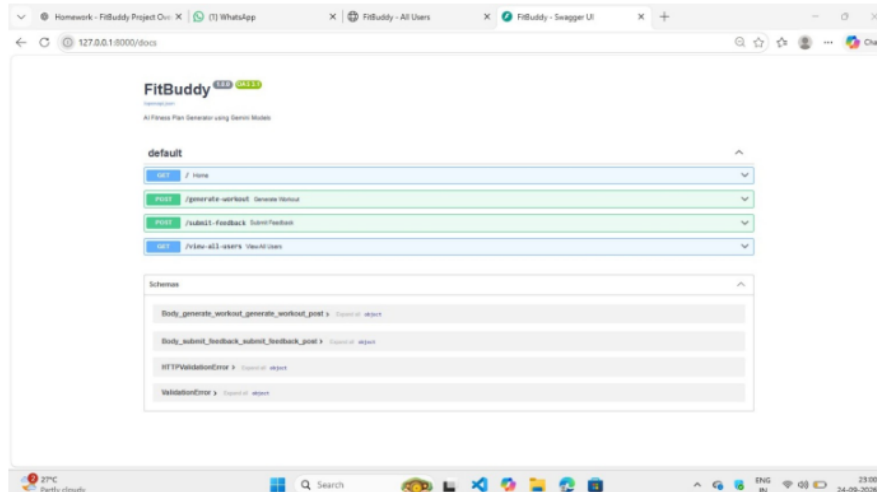


Figure 2: FitBuddy Swagger UI showing available API endpoints.

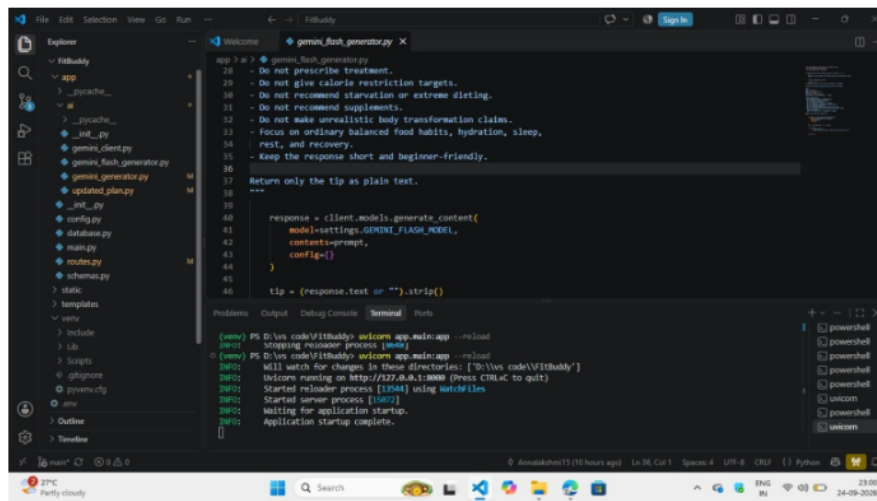


Figure 3: FitBuddy generated 7-day fitness plan.

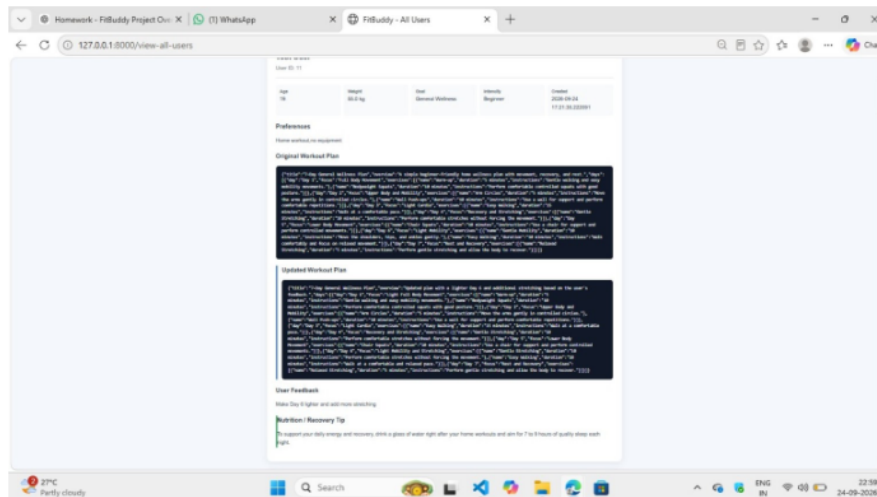


Figure 4: Workout plan with Day 4, Day 5, Day 6 and Day 7 activities.

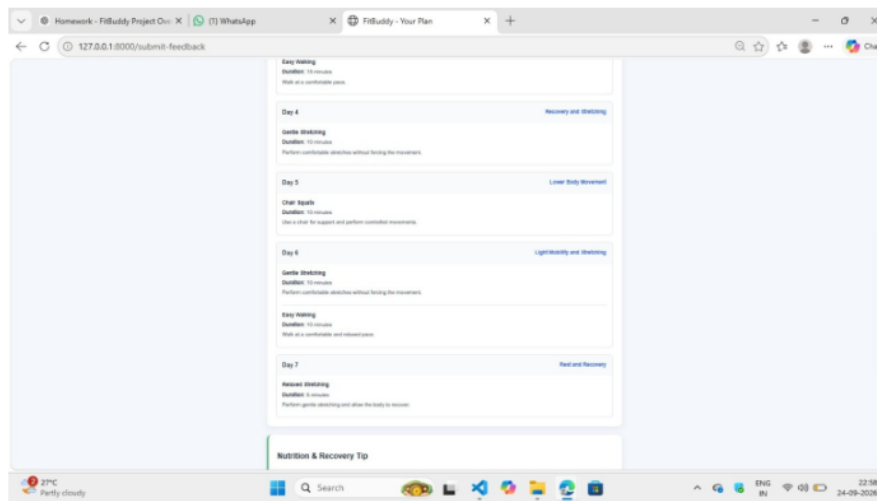


Figure 5: Feedback section for improving the generated fitness plan.

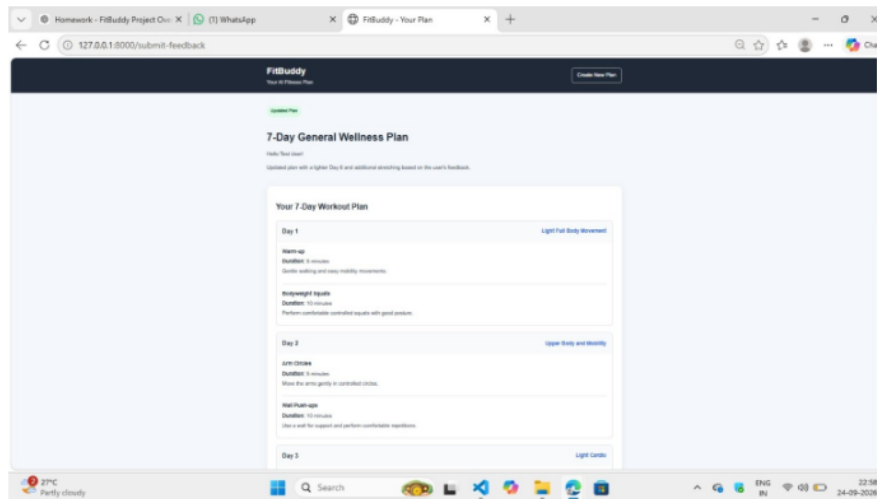


Figure 6: Updated FitBuddy plan after user feedback.

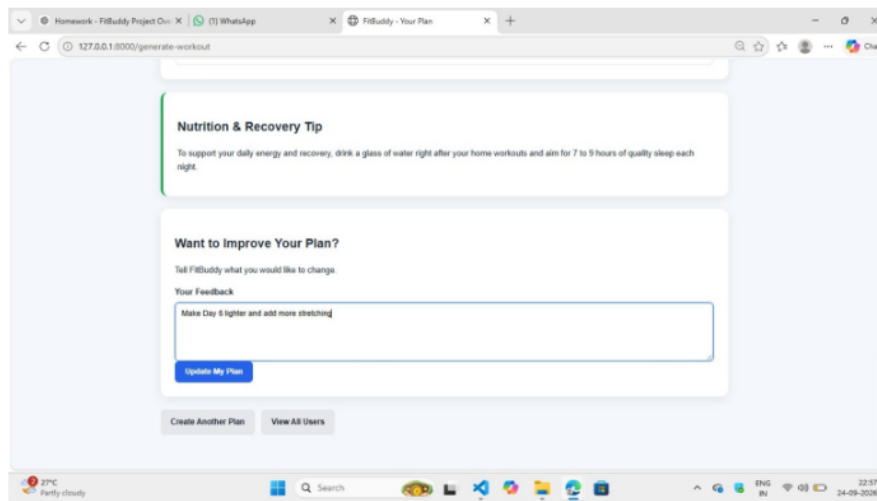


Figure 7: View All Users page showing stored workout-plan information.

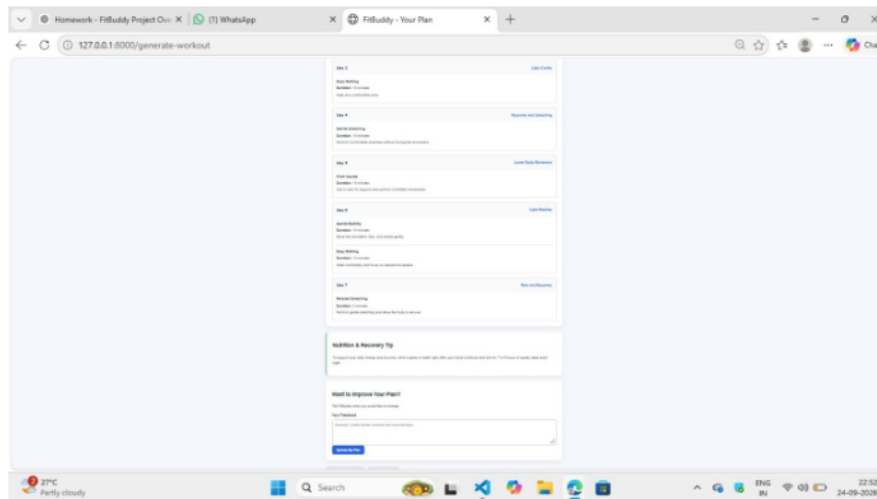


Figure 8: Nutrition and Recovery Tip section.

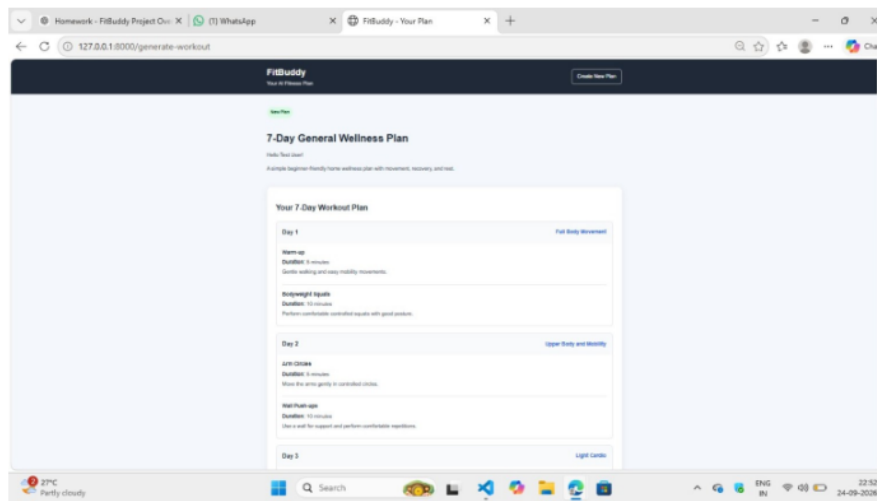


Figure 9: Additional FitBuddy workout-plan output screen.